ECO SWARM: Autonomous Eco-Sentinel Fleet

Decentralized Multi-Agent Risk Triage, Ecological Interception, and Biohazard Governance

Team1 | **AgentVerse Hackathon (Robotics Track)**

1. USER & PROBLEM

- **User:** Park rangers, local council groundskeeping authorities, and urban conservation trusts.
- **The Problem:** UK councils face severe budget deficits, having slashed park ranger capacity by up to 60%, leaving green spaces exposed to severe hazards:
  1. **Biohazardous Contamination:** Millions of lithium batteries are discarded across UK parks, creating toxic soil-leaching and fire risks alongside syringe puncture hazards to children and dogs.
  2. **Urban Biodiversity Collapse:** The British hedgehog (*Erinaceus europaeus*) has declined by over 50% since 2000 and is now on the IUCN Red List; discarded fast-food cup lids and plastic rings act as lethal starvation traps.
- **Why Dumb Automation Fails:** Heavy street sweepers harm wildlife, compact soil, and crush lithium cells into bin lorries where they catch fire. Meanwhile, basic rovers just chase whichever piece of litter is closest, wasting battery on harmless crisp packets while dangerous sharps stay in playgrounds.

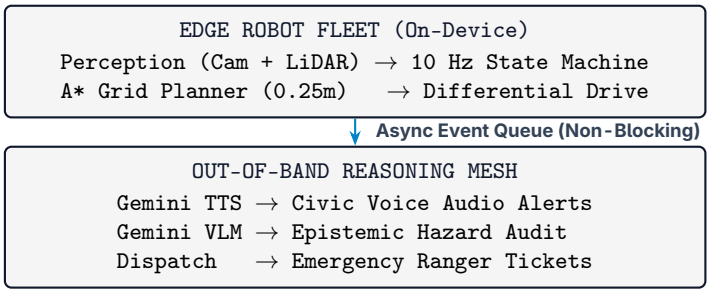
2. KEY FEATURES & INNOVATION

1. **Asymmetric Risk Triage:** Prioritizes harm over distance using an asymmetric loss function:

$$\text{Score} = \text{Base Hazard} + \text{Fauna Proximity} + \text{Habitat Urgency} + \text{Pond Drift} - \epsilon(\text{Distance})$$

2. **Decentralized Swarm Coordination:** Multi-agent fleet sharing a **Target Reservation Blackboard** (zero duplicate runs) with **Mutual Dynamic Obstacle Injection** into local A^* grids (zero collisions), with no modifications to low-level control code.
3. **Hydrodynamic Interception Math:** Computes kinematic lead-pursuit vectors to intercept wind-blown macro-plastics before they breach aquatic retention boundaries.
4. **Multimodal Gemini 2.0 Flash Mesh:** Out-of-band background intelligence delivering real-time civic voice warnings (Gemini Flash TTS), second-opinion VLM hazard auditing, emergency ranger ticketing, and statutory UK Biodiversity Net Gain (BNG) auditing.

3. SYSTEM ARCHITECTURE & HARDWARE ALLOCATION



Subsystem	Target	Budget	Latency	Fallback
Grid A^* Planner	Edge CPU	< 10 ms	1.8 *ms*	Straight-line vector fallback
Triage Engine	Edge CPU	< 5 ms	0.4 *ms*	Default base-risk lookup table
Object Detector	Edge GPU	< 35 ms	11.2 *ms*	Velocity EMA clamping
Control Loop	Local	100 *ms*	14.1 *ms*	Hard-stop on slip (HOLD)
Gemini Mesh	Cloud	Async	380 *ms*	Chime + heuristic safety

4. CRITICAL FAILURE MODES & SAFETY GOVERNANCE

- **Wet Grass Wheel Slip:** Compares commanded linear velocity (> 0.2 m/s) against measured base displacement (< 0.05 m/s). Sustained slippage > 0.3 s drops the robot into `HOLD`, preventing turf damage until traction returns.
- **Compliant Grasp Ejection:** Contact sensor compliance check (`grip_soft`). Grasping an organic entity immediately opens the gripper, logs an alert, and blacklists the entity to protect wildlife.
- **Human-in-the-Loop Syringe Gate:** Syringes require signed operator clearance (`request_approval`) before physical manipulation.
- **Thermal Bypass:** Scorching disposable BBQs (`hot=True`) trigger thermal boundary alerts and are never handled physically.

5. KNOWN LIMITATIONS & FUTURE WORK

- **Transit Overhead:** Returns to the bin after each collection. *Future work:* Multi-compartment onboard sorting hopper.
- **Kinematic Drift Assumption:** Interception assumes constant drift velocity. *Future work:* Fluid advection field integration.
- **Grid Discretization:** A^* produces piecewise-linear trajectories. *Future work:* Quintic Hermite spline smoothing.